FinGuard Pulse

Cost-aware payment fraud decisioning with explainable alerts and drift monitoring

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Challenge track	Track 3: FinTech AI
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2. Problem Statement

Payment providers decide which transactions to hold using models that predict whether a payment is fraudulent, and judge those models on accuracy or AUC. That is the wrong yardstick. Fraud is rare (about 1 payment in 100 in our simulation), so a model can look accurate while stopping almost nothing, and the two kinds of mistake have very different prices: a missed fraud costs the whole amount, while a false alarm costs a customer's time and sometimes the sale. The business question is: **for which payments does holding the transaction save more money than it costs?**

Problem statement: Build an AI system that scores every digital payment for fraud risk, decides which payments to flag by comparing expected rupees saved with expected cost of friction, explains what drives the risk, and keeps working as fraud tactics change.

3. Problem Understanding

Background

Digital payments such as UPI, cards and wallets are used for everything from tea to rent, and fraud on them is rare but expensive. Standard fraud models output a probability and apply one fixed cut-off for every payment, usually 0.5. With heavily imbalanced data, that cut-off is almost never reached, so the model flags very little. Fraud tactics also move: account takeover and large out-of-pattern payments give way to social-engineering scams, where the victim approves the payment themselves, so a model that was accurate at launch quietly degrades.

Four outcomes of every decision

Outcome	What happens	Cost or value (our economics)
Fraud caught	Payment flagged and the fraud is real	Saves the full amount, minus the cost of the flag
Fraud missed	Payment approved but it is fraud	Loses the full amount (the largest cost)
False alarm	Genuine payment flagged	INR 80 friction + INR 10 handling
Genuine approved	Nothing happens	No cost; the correct outcome for about 99% of payments

Personas and pain points

- **Fraud operations analyst:** works through a queue of alerts, most of which are false, and has no way to see which alerts are worth the time.
- **Head of risk:** must balance fraud losses against customer friction but cannot show the trade-off in rupees.
- **Payments product manager:** every wrongly declined payment is a lost sale and a frustrated customer.
- **Customer:** either loses savings to a scam, or has a genuine payment blocked at the worst moment.

Impact

Commercially, missed fraud is a direct loss and false alarms erode conversion and trust. Socially, first-time digital users and less experienced users are commonly targeted by social-engineering scams, so catching these early matters most for the people least able to absorb the loss.

4. Proposed Solution

FinGuard Pulse is a decision system, not only a model. It has four parts:

- **Risk engine.** A gradient-boosted classifier trained on labelled payment history outputs the probability that each payment is fraud. A logistic regression is kept as a simple benchmark.
- **Expected-value decision rule.** A payment is flagged when the expected loss prevented is larger than the expected cost of flagging: **p x amount > (1 - p) x friction + ops cost**. The bar moves with the amount, so a Rs 40,000 payment is flagged at a much lower risk score than a Rs 300 one. The cost figures are inputs the risk team controls.
- **Explainability.** Permutation importance shows which signals drive the model (new device, new payee, amount against the customer's usual). Per-transaction reason codes for analysts are planned.
- **Drift monitor.** The Population Stability Index (PSI) is computed per feature against the training data; above 0.25 is treated as a major shift that calls for retraining. The prototype demonstrates detection and measures the benefit recovered by retraining. Automated alerts and scheduled retraining are planned.

How a risk analyst uses it

The analyst sets the friction and handling cost. The dashboard shows net benefit for each policy, scores a single transaction on demand (decision, risk score and the signals behind it), and shows the drift status of the incoming data.

5. Innovation and Uniqueness

- **Optimises net rupees saved, not accuracy.** Every policy is scored in INR on held-out payments, not by AUC.
- **Amount-aware decisions.** One global cut-off treats a Rs 300 and a Rs 40,000 payment alike; the expected-value rule does not.
- **Shows the failure of the usual setup.** The default 0.5 cut-off is compared with a tuned flat cut-off and the expected-value rule on identical data, so the benefit of each step is measured.
- **Drift is part of the product.** A shift in fraud tactics triggers an alert and a measured recovery step, instead of a silent decline.

Compared with existing alternatives

Approach	Limitation addressed by FinGuard Pulse
Hand-written rules (amount limits, device checks)	Static and narrow; catches only the patterns someone already thought of
ML score with default 0.5 cut-off	On rare fraud almost nothing crosses the cut-off, so little is stopped (Section 8)
ML score with one tuned flat cut-off	Better, but ignores the amount at stake; misses cheap-to-catch large frauds
Flagging or approving everything	Approving all loses the full fraud value; flagging all makes every genuine customer pay the friction cost

6. Solution Architecture

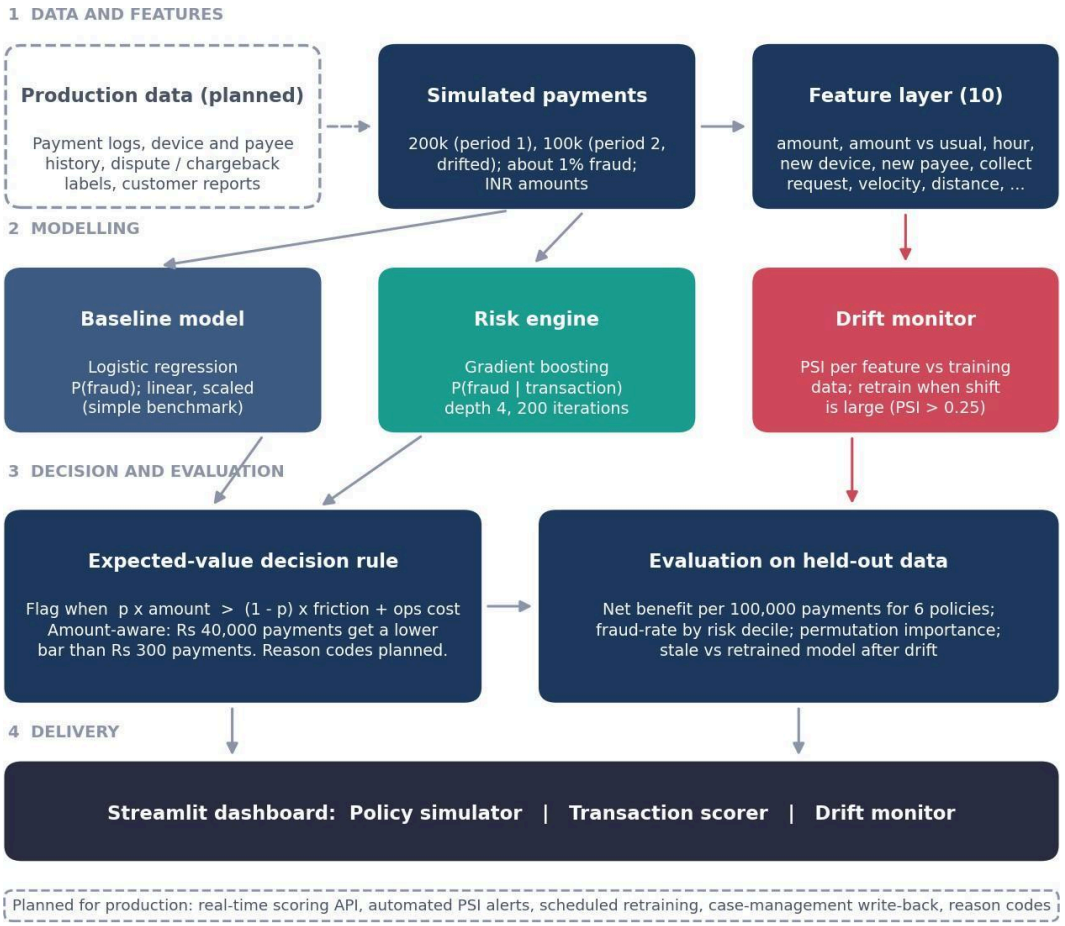


Figure 1. System architecture. Solid boxes are implemented in the prototype; dashed boxes are planned production components.

Workflow

- **Data:** the prototype generates simulated payments with fraud labels. In production this is replaced by payment logs, device and payee history, and dispute or chargeback labels.
- **Features:** amount, amount against the customer's usual, hour of day, new device, new payee, collect-request flag, payments in the last hour, distance from usual location, account age, failed PIN attempts in 24 hours.
- **Models:** a logistic regression (benchmark) and a gradient-boosting classifier, trained on period-1 data.
- **Decide:** flag each payment with the expected-value rule, using the model probability and the payment amount.
- **Evaluate:** net benefit per 100,000 payments for six policies on held-out data, plus fraud rate by risk decile.
- **Monitor:** PSI of period-2 features against period-1 training data, then compare the stale model with one retrained on period-2 data.

Technology stack

Layer	Technology
Data and features	Python, pandas, NumPy
Modelling	scikit-learn HistGradientBoostingClassifier and LogisticRegression
Evaluation and monitoring	Custom net-benefit estimator, PSI, permutation importance
Dashboard	Streamlit
Reporting	Matplotlib, ReportLab

7. AI and Technology Used

Component	Detail
Dataset	Simulated payments. Period 1: 200,000 payments (120,000 train, 40,000 validation, 40,000 test), fraud rate 1.00%. Period 2 with drift: 100,000 payments (50,000 retrain, 50,000 test), fraud rate 1.23%.
Features (10)	amount, amount_ratio, hour, new_device, new_payee, collect_request, txns_last_hour, geo_distance_km, account_age_days, failed_pin_24h
Risk engine	HistGradientBoostingClassifier: depth 4, 200 iterations, learning rate 0.06, L2 regularisation 1.0; predicts P(fraud)
Benchmark model	Logistic regression on standardised features
Decision rule	Flag when p x amount > (1 - p) x INR 80 + INR 10. Compared with a flat score cut-off chosen on the validation set.
Economics (assumptions)	Flagged fraud is stopped and its full amount saved. Every flag costs INR 10 to handle; each flagged genuine payment costs a further INR 80 in friction and abandonment.
Evaluation	Net benefit = amount saved on flagged fraud less flag costs, scaled to 100,000 payments. 95% intervals from 400 bootstrap resamples of the test set.
Drift	PSI with 10 quantile bins against period-1 training data (binary features compared by category)

The data is simulated because real labelled payment data is confidential. The simulation was built with two fraud patterns: in period 1, account takeover (new device, new payee, large out-of-pattern amounts, night hours); in period 2, collect-request scams with smaller amounts, daytime activity and the victim's own device. Section 10 explains what this means for how far the results can be trusted.

8. Prototype and Model Results

The Streamlit prototype (`app.py`) has three tabs: a policy simulator, a single-transaction scorer and a drift monitor. The figures below are generated by the same code the prototype uses (`finguard.py`).

Result 1: which policy earns the most

Policy (40,000 held-out payments)	Net benefit per 100,000 (INR)	Fraud caught	Fraud value caught	False alarms per 100,000
Approve everything (no screening)	0	0.0%	0.0%	0
Hand-written rules	260,132	2.9%	13.3%	32
Gradient boosting, default 0.5 cut-off	319,592	7.8%	16.3%	30
Gradient boosting, tuned flat cut-off (0.059)	963,908	51.6%	60.7%	2,620
Logistic regression + expected-value rule	1,080,715	47.7%	69.2%	3,200
Gradient boosting + expected-value rule	1,108,060	48.7%	70.0%	3,078

The default 0.5 cut-off flags only 105 payments per 100,000 and catches 7.8% of fraud, because a model on 1%-fraud data rarely outputs a probability that high. Tuning one flat cut-off on a validation set (to 0.059) lifts net benefit to INR 963,908. The expected-value rule reaches INR 1,108,060, because it lets cheap-to-catch large payments through at lower risk scores. The gap between the rule and the tuned flat cut-off is INR 144,152 per 100,000 (95% bootstrap interval INR 27,024 to 290,140), so it is unlikely to be chance, although the test set contains only 384 fraud cases.

One honest finding: swapping gradient boosting for logistic regression under the same rule gives INR 1,080,715, only 3% less. Most of the gain in this simulation comes from the decision rule, not from the more complex model. Model quality for reference: ROC-AUC 0.91 and PR-AUC 0.25 for gradient boosting, against 0.90 and 0.21 for logistic regression.

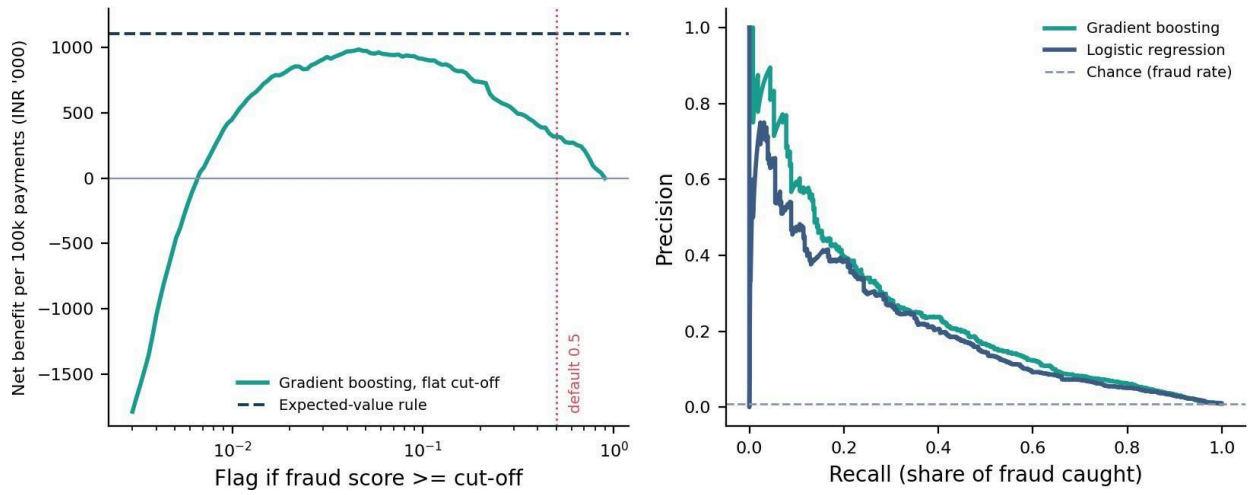


Figure 2. Left: net benefit of a flat cut-off as it varies, against the expected-value rule (dashed). Right: precision-recall curves on held-out data.

Result 2: does the ranking hold up?

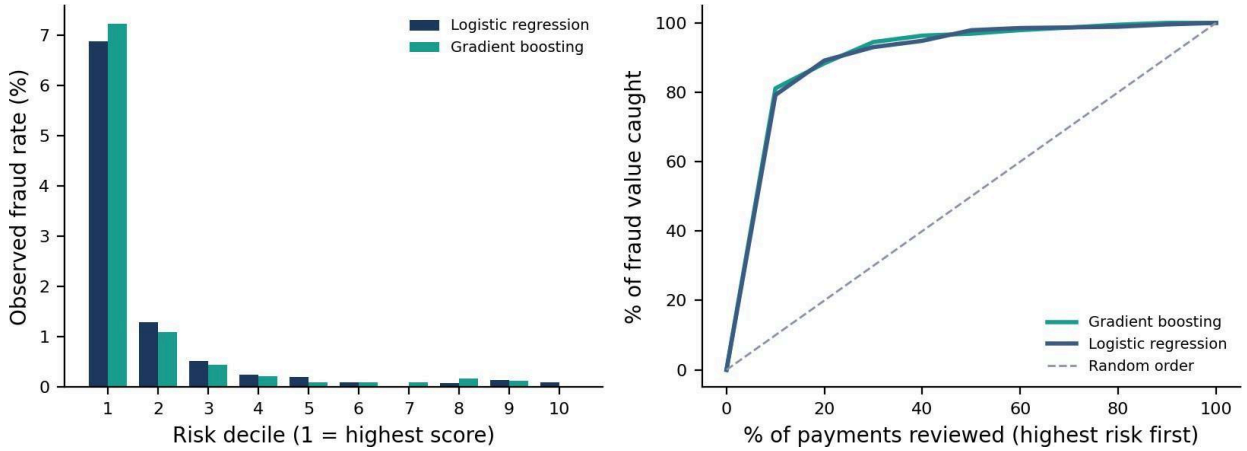


Figure 3. Left: observed fraud rate by risk decile on held-out data. Right: cumulative share of fraud value caught as more payments are reviewed.

The top risk decile under gradient boosting has an observed fraud rate of 7.2%, against 1.0% overall, and holds 75% of all fraud cases and 81% of fraud value. By the fifth decile the fraud rate is about 0.1%. Reviewing the riskiest 10% of payments therefore covers most of the loss, which is what makes a small analyst team workable.

Result 3: drift

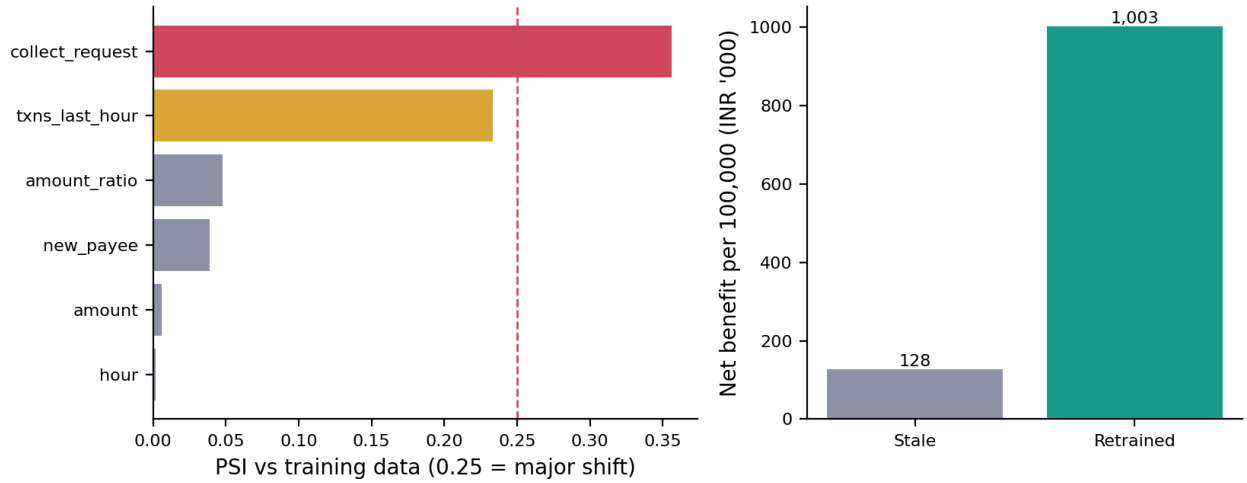


Figure 4. Left: feature drift between period 1 and period 2 (red above 0.25, amber 0.10 to 0.25). Right: net benefit of the stale model against a model retrained on period-2 data.

In period 2 collect-request payments rise from 8% to 30% of traffic, and the pattern of fraud changes. The monitor flags **collect_request** (PSI 0.36, above 0.25) and shows **txns_last_hour** at 0.23, close to the threshold. The stale model's ROC-AUC falls from 0.91 to 0.83, which may still look acceptable, but its PR-AUC falls from 0.25 to 0.06, and net benefit collapses to INR 127,530 per 100,000. Retraining on period-2 data restores it to INR 1,002,766, about 7.9 times higher. The hand-written rules, built for period-1 patterns, earn INR 27,066 in period 2.

9. Business Impact

All figures are from the simulation under the stated economics. The gaps between policies are the point; real values depend on actual fraud rates, ticket sizes and costs.

Metric	Result in simulation (per 100,000 payments)	Meaning
Net benefit vs no screening	INR 1,108,060 better	Amount saved, after paying all flag costs; 70% of fraud value caught
Net benefit vs hand-written rules	INR 847,928 better	Value of a learned, probability-based approach
Net benefit vs default 0.5 cut-off	INR 788,468 better	Value of choosing the cut-off on cost, not by habit
Net benefit vs tuned flat cut-off	INR 144,152 better	Value of the amount-aware expected-value rule alone
Retraining after drift	INR 875,236 recovered	Value of monitoring and retraining

Illustrative scaling

For a provider handling 5,000,000 payments a month, the simulated gap between the expected-value rule and a tuned flat cut-off would be about INR 7,207,625 per month, and the gap to the default 0.5 cut-off about INR 39,423,375 per month. This is an extrapolation of simulated numbers and should be read as an order of magnitude to be validated in a shadow-mode pilot, not as a forecast.

Customer experience

The expected-value rule flags about 3,545 of every 100,000 payments (3.5%), so more than 96% of customers never see any friction. It does create 3,078 false alarms per 100,000, and that is a real cost the risk team can tune by raising the friction-cost input. Genuine payments flagged should be resolved with a quick step-up check rather than a hard decline.

10. Feasibility

Requirements

- Labelled payment history with fraud outcomes (disputes, chargebacks, confirmed customer reports) and logged device, payee and timing information.
- A fraud analyst, a data analyst and a risk owner who agrees the cost inputs. No GPUs are needed; scoring 10,000 payments took about 49 ms on a laptop CPU in our test (batch scoring, excluding data lookups).
- A way to record which payments were flagged and what happened next.

Technical challenges and honest limitations

- **Simulated data.** The strong results partly reflect that the simulation was built with clear fraud patterns. Real fraud is noisier and the first real-data pilot is the true test.
- **Rare events and label delay.** Fraud outcomes can take days or weeks to confirm, so recent data is under-labelled and retraining must allow for it.
- **Adversaries adapt.** Once a pattern is blocked, fraudsters change it; drift monitoring is a safeguard, not a guarantee.
- **Cost inputs are assumptions.** The INR 80 friction cost needs validating with finance and customer-support data; the ranking of policies depends on it.
- **Complex model, modest gain.** In our results the simple model comes close under the same rule. A production team should benchmark both before choosing.
- **Privacy, fairness and regulation.** Personal data must be handled under India's Digital Personal Data Protection Act, 2023, and the system must meet the payment-security and grievance rules that apply to the deploying institution. Flag rates should be audited across customer groups.

Scalability plan

Phase	Scope	Success test
1. Shadow pilot (4 to 6 weeks)	Score live or replayed payments without acting on the scores	Positive net benefit against current rules on confirmed fraud
2. Limited live	Flag a share of traffic; keep a random unflagged sample to measure missed fraud	Stable decile fraud rates over 3 months
3. Platform	Real-time API, scheduled retraining on drift alerts, case-management integration	Lower fraud loss at equal or lower false-alarm rate

11. Future Scope

- **Three-way action:** approve, step-up authentication, or block, each with its own cost, instead of a single flag.
- **Per-transaction reason codes:** show the analyst why a payment scored high (for example, new payee plus a collect request at an unusual amount).
- **Network features:** link payees and devices in a graph to find mule accounts that no single transaction reveals.
- **Sequence and behaviour models:** use the order of a customer's recent actions, not only single-payment features.
- **Analyst feedback loop:** use confirmed and dismissed alerts to retrain and to prioritise which cases to label first.
- **Adversarial testing:** simulate new scam patterns to check the monitor and model before real fraudsters do.
- **Generative AI layer:** draft case summaries and customer messages for flagged payments, with human approval.
- **Fairness checks and integration:** audit flag rates across groups; connectors for payment gateways and case-management tools.

12. Conclusion

Predicting fraud is not the same as deciding what to do about it. FinGuard Pulse turns a fraud probability into a decision by weighing the rupees at stake in each payment against the cost of stopping it, explains what drives the risk, and watches for drift so that performance does not silently decay.

On a simulated payment stream, approving everything lost the full fraud value, a default 0.5 cut-off earned INR 319,592 per 100,000 payments, and the expected-value rule earned INR 1,108,060 while flagging 3,545 payments. When fraud tactics shifted, the drift monitor flagged the change and retraining recovered most of the lost value. The next step is a shadow-mode pilot on real payment data, where the central claim of this report can be tested.

Key value propositions

- Optimises net rupees saved, not prediction accuracy.
- Amount-aware decisions: effort goes where the money is.
- Explainable signals and a tunable cost trade-off for risk teams.
- Built-in drift monitoring; light enough to run on a laptop or a free hosting tier.